Form C

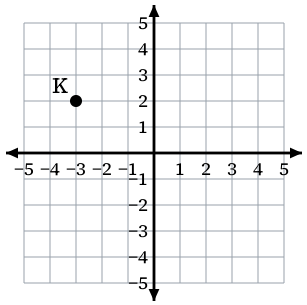
Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

Name _____ Period _____ Date _____

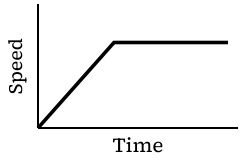
Part I – Selected Response

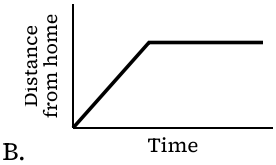
1. Which ordered pair represents the point graphed?

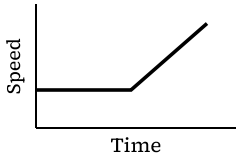


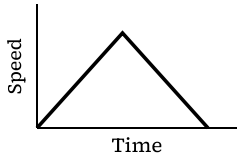
A. (3, −2)
B. (2, −3)
C. (−3, 2)
D. (−2, 3)

2. Which graph shows a car speeding up and then driving at a constant speed?

A. 

B. 

C. 

D. 

3. If the values in the table below model direct variation, what is the rate?

x	0	1	2	3	4
y	0	7	14	21	28

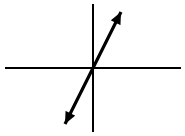
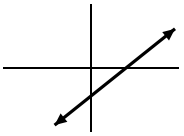
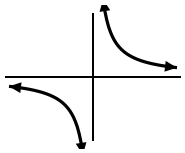
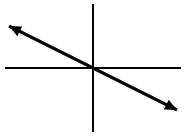
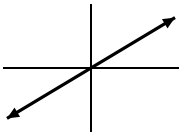
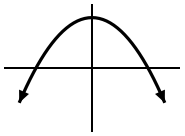
A. $\frac{1}{7}$
B. 14
C. 7
D. It does not model direct variation.

4. Which equation is the direct variation equation for the values in the table below?

x	0	2	4	6	8
y	0	10	20	30	40

A. $y = 10x$
B. $y = 5x$
C. $y = 2x$
D. $y = \frac{1}{5}x$

5. Which graphs show direct variation? Circle all that apply.

A. 
B. 
C. 
D. 
E. 
F. 

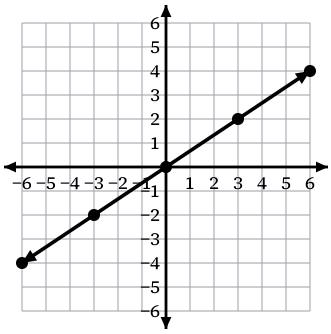
6. Which equations are direct variation equations? Circle all that apply.

A. $y = x - 5$
B. $y = 9x$
C. $y = \frac{10}{x}$
D. $y = -\frac{3}{4}x$
E. $y = 2^x$
F. $y = \frac{x}{3}$

7. What is the slope of the direct variation equation $y = 0.6x$?

A. 6
B. $\frac{3}{5}$
C. $\frac{1}{6}$
D. $\frac{5}{3}$

8. What is the direct variation equation for the line graphed below?



A. $y = \frac{3}{2}x$

B. $y = -\frac{2}{3}x$

C. $y = 2x$

D. $y = \frac{2}{3}x$

Form C

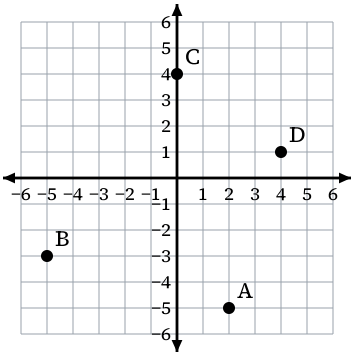
Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

Name _____ Period _____ Date _____

Part II – Constructed Response

Write the ordered pair for each point on the coordinate plane to the right.



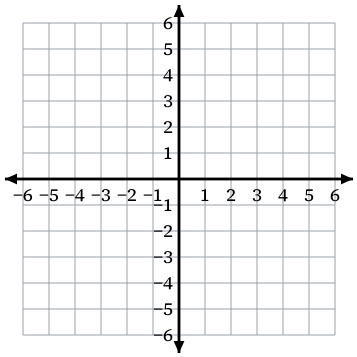
1. A

2. B

3. C

4. D

5. Plot the points (−4, 3), (4, 3) and (0, −3) and connect them in order. Connect the last point listed to the first point. What shape do the points create?



Shape: _____

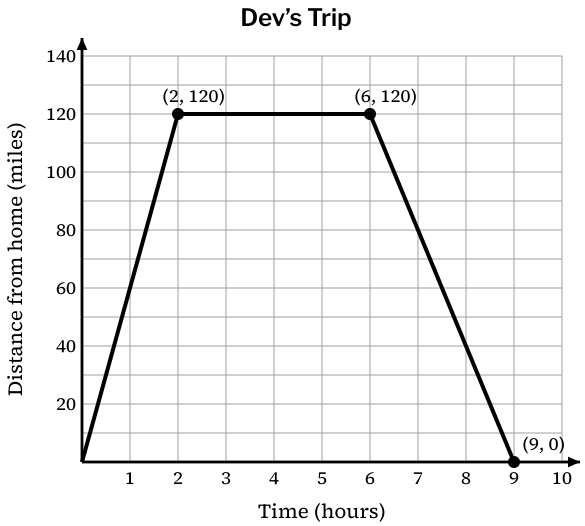
6. Dev's family drove to a lake, spent time there, and then drove back home. The graph below shows their distance from home (miles) over time (hours). For each part, explain how you know your answer is correct.

a. How far away is the lake from Dev's house?

b. How long did the family stay at the lake?

c. Find the family's rate of speed in miles per hour for the drive to the lake.

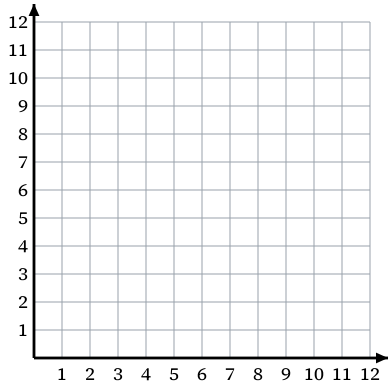
d. Find the family's rate of speed in miles per hour for the drive home.



Complete each input-output table and graph each function in Quadrant I. What is the slope of each direct variation equation?

7.

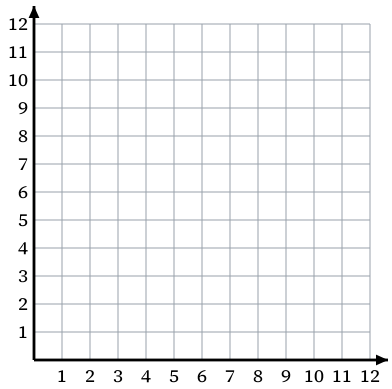
Input x	Function Rule $y = \frac{1}{3}x$	Output y
0		
3		
6		
9		
12		



slope: _____

8.

Input x	Function Rule $y = \frac{3}{2}x$	Output y
0		
2		
4		
6		
8		



slope: _____

Determine whether each table models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

9.

x	y
−4	−2
−2	−1
0	0
2	1
4	2

direct variation? **YES** **NO**

rate/explanation:

10.

x	y
−2	−10
−1	−5
0	0
1	5
2	12

direct variation? **YES** **NO**

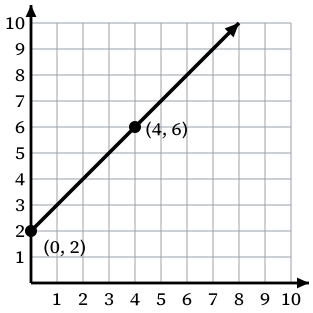
rate/explanation:

Determine whether each equation models direct variation. Circle YES or NO. If it does, give the rate. If not, explain why not.

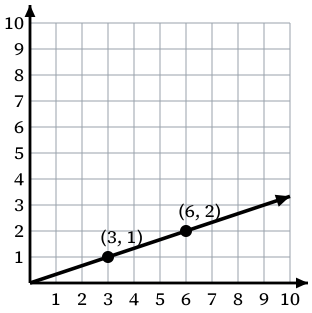
11. $y = 6x^2$ direct variation? **YES** **NO** rate/explanation:

12. $y = \frac{x}{5}$ direct variation? **YES** **NO** rate/explanation:

13. Ken says Graph A models direct variation because it is a straight line and has a rate of 1. Rosa says Graph B models direct variation and has a rate of $\frac{1}{3}$. Which person is correct? Explain your reasoning.



Graph A

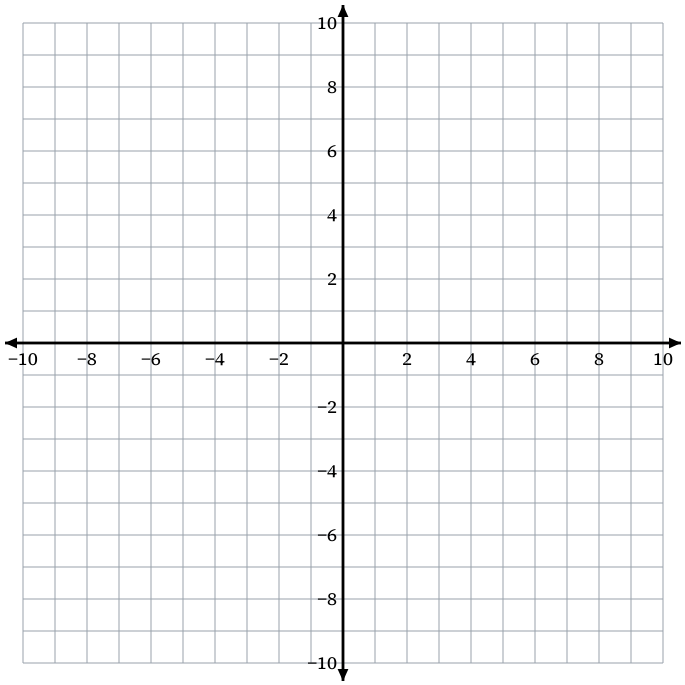


Graph B

14. Tickets to the school play cost $6 each. The table below shows the total cost for the number of tickets bought. Write an equation that shows the total cost, y, of x tickets.

x	y
0	0
3	18
6	36
9	54
12	72

15. Graph the direct variation equation y = 3x.



Core Focus on Proportions & Probability

Block 5 Review ~ Direct Variation

Name _____ Period _____ Date _____

Part III – Problem Solving

The table below shows the amount of money Marcus earned walking dogs in his neighborhood.

Hours Walking Dogs	Money Earned ($)
0	0.00
2	23.00
4	46.00
6	69.00
8	92.00

How many hours will Marcus need to walk dogs to earn $161.00?